

INTERNET OF THINGS (IoT)

Professional Training Curriculum

A structured curriculum covering IoT fundamentals, electronics, Arduino, embedded systems, ESP32/NodeMCU, Raspberry Pi, networking, cloud platforms, application development, security, edge computing, and a live IoT project.

PROGRAM STRUCTURE

12 Comprehensive Modules

IoT SYLLABUS

Module 1 — IoT Fundamentals

- IoT Concepts & Architecture
- IoT Components & Applications
- Sensors, Actuators & Controllers
- IoT Ecosystem & Use Cases

Module 2 — Electronics for IoT

- Basic Electronics & Circuit Fundamentals
- Digital & Analog Signals
- Sensors & Actuators Interfacing
- Breadboard, Components & Circuit Design

Module 3 — Arduino Programming

- Arduino Architecture & IDE
- C/C++ Programming Fundamentals
- Digital & Analog I/O
- Sensors & Actuators
- Serial Communication

Module 4 — Microcontrollers & Embedded Systems

- Microcontrollers & Embedded Systems
- GPIO, PWM, ADC & Interrupts
- Embedded Programming
- Hardware-Software Interfacing

Module 5 — ESP32/NodeMCU & Wireless IoT

- ESP32/NodeMCU Architecture
- Wi-Fi & Bluetooth Connectivity
- Sensor Data Acquisition
- Wireless IoT Applications

Module 6 — Raspberry Pi & Python

- Raspberry Pi Setup & Linux
- Python Programming for IoT
- GPIO & Hardware Interfacing
- Raspberry Pi IoT Applications

Module 7 — IoT Networking & Protocols

- TCP/IP & IoT Networking
- MQTT & HTTP/HTTPS
- CoAP & Communication Models
- Device-to-Device Communication
- Device-to-Cloud Communication

Module 8 — IoT Cloud & Data Management

- IoT Cloud Architecture
- Device Registration & Management
- Cloud Data Storage & Processing
- IoT Dashboards & Visualization

Module 9 — IoT Application Development

- Real-Time Sensor Monitoring
- Web/Mobile IoT Interfaces
- APIs & Database Integration
- Remote Device Control

Module 10 — IoT Security

- IoT Security Fundamentals
- Device & Network Security
- Authentication & Encryption
- IoT Vulnerability & Security Testing

Module 11 — Advanced IoT & Edge Computing

- Edge & Fog Computing
- IoT Data Analytics
- AI/ML for IoT
- Smart Home IoT
- Smart Agriculture
- Industrial IoT

Module 12 — Live IoT Project

- IoT Problem Identification
- System Design
- Hardware & Sensor Integration
- Connectivity Integration
- Cloud Dashboard
- Real-Time Monitoring
- Final IoT Product Demonstration

CORE TECHNOLOGIES

Arduino • ESP32/NodeMCU • Raspberry Pi • Python • C/C++ • MQTT • Sensors • Actuators • APIs •
Databases • IoT Cloud Platforms

Program Outcome: Learners develop practical skills to design, build, connect, secure, monitor, and deploy IoT systems, culminating in a real-world IoT product demonstration.